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14.7 Information-Theoretic Properties of Distributions

In this section we return to nominal distributions, that is to say, to distributions with discrete outcomes that have no meaningful ordering. Information theory [1-3] provides a different, and sometimes very useful, perspective on the nature of such a distribution $\mathbf{p}$ with outcomes i , $0 \leq i \leq I - 1$, and associated probabilities p_i , and on the relation between two or more such distributions. We develop that perspective in this section, starting with a review of some key concepts.

14.7.1 Entropy of a Distribution

Suppose that we make M sequential, independent draws from a distribution $\mathbf{p}$, thus generating a *message* that describes the outcomes, an M -vector of integers i_j , each in the range $0 \leq i_j \leq I - 1$, with $j = 0, \dots, M - 1$. We want to send the message to a waiting confederate, but we first want to compress it (that is, *encode* it) into the most parsimonious format, say into the smallest possible number of bits, B . We can calculate a lower bound on B by equating 2^B , the number of possible different compressed messages, to a statistical estimate of the number of likely input messages. That equation, in the limit of M becoming very large, is

$$2^B \approx \frac{M!}{\prod_i (Mp_i)!} \quad (14.7.1)$$

The rationale for the right-hand side is that our message will contain very nearly Mp_i occurrences of the integer i for each i , so the count of messages will be very nearly the number of ways that we can arrange M objects of I types, with Mp_i of them identical for each type i . Taking the logarithm of equation (14.7.1), using Stirling's approximation on the factorials, and keeping only terms that scale as fast as M , we readily obtain

$$B \approx -M \sum_{i=0}^{I-1} p_i \log_2 p_i \equiv M H_2(\mathbf{p}) \quad (14.7.2)$$

where $H_2(\mathbf{p})$ is called the *entropy (in bits)* of the distribution $\mathbf{p}$, a terminology borrowed from statistical physics. The subscript 2 is to remind us that the logarithm has

base 2. We can also define an entropy with base e ,

$$H(\mathbf{p}) \equiv - \sum_{i=0}^{I-1} p_i \ln p_i = -(\ln 2) \sum_{i=0}^{I-1} p_i \log_2 p_i = (\ln 2) H_2(\mathbf{p}) \quad (14.7.3)$$

If $H_2(\mathbf{p})$ is measured in *bits*, then $H(\mathbf{p})$ will be measured in *nats*, with 1 nat = 1.4427 bits. In evaluating (14.7.3), note that

$$\lim_{p \rightarrow 0} p \ln p = 0 \quad (14.7.4)$$

The value $H(\mathbf{p})$ lies between 0 and $\ln I$. It is zero only when one of the p_i 's is one, all the others zero.

Although we derived B as a lower bound, a central result of information theory is that, in the limit of large M , one can find codes that actually achieve that bound. (Arithmetic coding, described in §22.6, is an example of such a code.) Heuristically, one can interpret equation (14.7.2) as saying that it takes, on average, $-\log_2 p_i$ bits (a positive number, since $p_i < 1$) to encode an outcome i . Thus, the compressed message size is M times the expectation of $-\log_2 p_i$ over outcomes occurring with probability p_i .

Yet a different view of entropy arises if we consider the game of “twenty questions,” where by repeated yes/no questions you try to eliminate all except one correct possibility for an unknown object. Better yet, let us consider a generalization of the game, where you are allowed to ask multiple choice questions as well as binary (yes/no) ones. The categories in your multiple choice questions are supposed to be mutually exclusive and exhaustive (as are “yes” and “no”).

The value to you of an answer increases with the number of possibilities that it eliminates. More specifically, an answer that eliminates all except a fraction p of the remaining possibilities can be assigned a value $-\ln p$. The purpose of the logarithm is to make the value additive, since, e.g., one question that eliminates all but 1/6 of the possibilities is considered as good as two questions that, in sequence, reduce the number by factors 1/2 and 1/3.

So that is the value of an answer; but what is the value of a question? If there are I possible answers to the question and the fraction of possibilities consistent with answer i is p_i , then the value of the question is the expectation value of the value of the answer, which is just $-\sum_i p_i \ln p_i$ or $H(\mathbf{p})$, as above.

As already mentioned, the entropy is zero only if one of the p_i 's is unity, with all the others zero. In this case, the question is valueless, since its answer is preordained. $H(\mathbf{p})$ takes on its maximum value when all the p_i 's are equal, in which case the question is sure to eliminate all but a fraction $1/I$ of the remaining possibilities.

A third, still different, view of entropy comes from thinking about bets (or, more politely, “investments”). A *fair bet* on an outcome i of probability p_i is one that has a payoff $o_i = 1/p_i$. This is the unique payoff (per unit wagered) for which, in the long run, the bettor will neither win nor lose, since in expectation value

$$\langle o_i \rangle = p_i o_i = 1 \quad (14.7.5)$$

Suppose you have the opportunity to bet repeatedly on a game offering fair bets on each outcome. This is not very interesting as a money-making proposition. But suppose that you are *clairvoyant* and can know in advance the outcome of each play

(although you cannot affect that outcome). Now you're in business! You always put your money on the winning choice of i . How much money can you make?

Since your profit on each (sure thing!) wager scales multiplicatively with your accumulated wealth, the appropriate figure of merit is the the average *doubling rate*, or, equivalently, *e-folding rate*, at which you can increase your capital. Since you always win, but can't control the outcome, this is given by

$$W \equiv \langle \ln o_i \rangle = \langle -\ln p_i \rangle = - \sum_i p_i \ln p_i = H(\mathbf{p}) \quad (14.7.6)$$

In other words, the entropy of a distribution is the e-folding rate of capital for a fair game about which you have perfect predictive information. While this may seem fanciful, we will see in §14.7.3 how it generalizes to the more realistic case where you have only imperfect, perhaps very small, predictive information.

14.7.2 Kullback-Leibler Distance

Back in the context of message compression, suppose that events occur with a distribution $\mathbf{p}$, that is, p_i , $0 \leq i \leq I - 1$, but we try to compress the message of their outcomes with a code that is optimized for some other distribution $\mathbf{q}$, that is, q_i , $0 \leq i \leq I - 1$. Our code therefore takes about $-\log_2 q_i$ bits, or $-\ln q_i$ nats, to encode outcome i , and the average compressed length per outcome is

$$- \sum_i p_i \ln q_i = H(\mathbf{p}) + \sum_i p_i \ln \frac{p_i}{q_i} \equiv H(\mathbf{p}) + D(\mathbf{p} \parallel \mathbf{q}) \quad (14.7.7)$$

The quantity

$$D(\mathbf{p} \parallel \mathbf{q}) \equiv \sum_i p_i \ln \frac{p_i}{q_i} \quad (14.7.8)$$

is called the *Kullback-Leibler distance* between $\mathbf{p}$ and $\mathbf{q}$, also called the *relative entropy* between the two distributions. We can easily prove that it is nonnegative, since

$$-D(\mathbf{p} \parallel \mathbf{q}) = \sum_i p_i \ln \left(\frac{q_i}{p_i} \right) \leq \sum_i p_i \left(\frac{q_i}{p_i} - 1 \right) = 1 - 1 = 0 \quad (14.7.9)$$

where the inequality follows from the fact that

$$\ln w \leq w - 1 \quad (14.7.10)$$

(Of course we already knew it had to be nonnegative, because we knew that $H(\mathbf{p})$ was the *smallest* possible compressed message size for the distribution $\mathbf{p}$.) The Kullback-Leibler distance between two distributions is zero only when the two distributions are identical. The Kullback-Leibler distance between any distribution $\mathbf{p}$ and the uniform distribution $\mathbf{U}$ is just the difference between the entropy of $\mathbf{p}$ and the maximum possible entropy $\ln I$, that is,

$$H(\mathbf{p}) + D(\mathbf{p} \parallel \mathbf{U}) = \ln I \quad (14.7.11)$$

This is illustrated in Figure 14.7.1. Just like entropy, the Kullback-Leibler distance is measured in bits or nats, depending on whether the logarithms are taken base 2 or e , respectively.

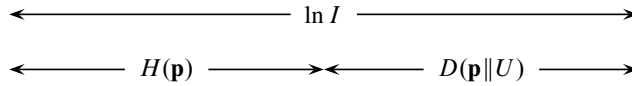


Figure 14.7.1. Relation between the entropy of a distribution $\mathbf{p}$, its Kullback-Leibler distance to the uniform distribution $\mathbf{U}$, and its maximum possible entropy $\ln I$.

Notice that the Kullback-Leibler distance is not symmetric, nor (it turns out) does it satisfy the triangle inequality. So it is not a true metric distance. It is, however, a useful measure of the degree by which some “target” distribution $\mathbf{q}$ differs from some “base” distribution $\mathbf{p}$. We now give a couple of examples of where it naturally occurs.

Example 1. Suppose that we are seeing events drawn from the distribution $\mathbf{p}$, but we want to rule out an alternative hypothesis that they are drawn from $\mathbf{q}$. We might do this by computing a likelihood ratio,

$$\mathcal{L} = \frac{p(\text{Data}|\mathbf{p})}{p(\text{Data}|\mathbf{q})} = \prod_{\text{data}} \frac{p_i}{q_i} \quad (14.7.12)$$

and rejecting the alternative hypothesis $\mathbf{q}$ if this ratio is larger than some large number, say 10^6 . (In the above shorthand notation, the product over “data” means that we substitute for i in each factor the particular outcome of that factor’s individual data event.) Taking the logarithm of equation (14.7.12), you can easily see that, under hypothesis $\mathbf{p}$, the average increase in $\ln \mathcal{L}$ per data event is just $D(\mathbf{p}||\mathbf{q})$. In other words, the Kullback-Leibler distance is the expected log-likelihood with which a false hypothesis $\mathbf{q}$ can be rejected, per event. As we might expect, this has something to do with “how different” $\mathbf{q}$ is from $\mathbf{p}$.

As a Bayesian aside, the reason that the above likelihood test is unsatisfyingly asymmetric is that, without the notion of a prior, we have no way to treat hypotheses $\mathbf{p}$ and $\mathbf{q}$ democratically. But suppose that $p(\mathbf{p})$ is the prior probability of $\mathbf{p}$, so that $p(\mathbf{q}) = 1 - p(\mathbf{p})$ is the prior for $\mathbf{q}$. Then the Bayes odds ratio on the two hypotheses is

$$\text{O.R.} = \frac{p(\mathbf{p}|\text{Data})}{p(\mathbf{q}|\text{Data})} = \frac{p(\text{Data}|\mathbf{p}) p(\mathbf{p})}{p(\text{Data}|\mathbf{q}) p(\mathbf{q})} = \frac{p(\mathbf{p})}{p(\mathbf{q})} \prod_{\text{data}} \frac{p_i}{q_i} \quad (14.7.13)$$

The figure of merit is now the expected increase in $\ln(\text{O.R.})$ if $\mathbf{p}$ is true, *minus* the expected increase (that is, *plus* the expected decrease) if $\mathbf{q}$ is true, which can readily be seen to be

$$p(\mathbf{p}) D(\mathbf{p}||\mathbf{q}) + p(\mathbf{q}) D(\mathbf{q}||\mathbf{p}) \quad (14.7.14)$$

per data event, which has the appropriate symmetry. We can use expression (14.7.14) to estimate how many data events we will need on average to distinguish between two distributions. Notice that in the case of a uniform (“noninformative”) prior, $p(\mathbf{p}) = p(\mathbf{q}) = 0.5$, we get just the symmetrized average of the two Kullback-Leibler distances.

Example 2. Meanwhile, back at the racetrack where we are offered payoffs of o_i on events with probability p_i , $\sum_i p_i = 1$, we want to work out the best way to divide our capital across all the possible outcomes i of each race. Suppose we bet a fraction b_i on outcome i . Analogously to equation (14.7.6), we want to maximize the

average e-folding rate,

$$W = \langle \ln(b_i o_i) \rangle = \sum_i p_i \ln(b_i o_i) \quad (14.7.15)$$

subject to the constraint

$$\sum_i b_i = 1 \quad (14.7.16)$$

An easy calculation (using a Lagrange multiplier to impose the constraint) gives the result that the maximum occurs for

$$b_i = p_i \quad (14.7.17)$$

completely independent of the values o_i ! This remarkable result is called *proportional betting*, or sometimes *Kelly's formula* [4].

In practice, the distribution $\mathbf{p}$ is imperfectly known, both to you and to the bookie at the track. Suppose that you estimate the outcome probabilities as $\mathbf{q}$, while the bookie's estimate is $\mathbf{r}$. If the bookie is feeling generous, he offers payoffs that are fair bets according to his estimate,

$$o_i = 1/r_i \quad (14.7.18)$$

while you place proportional bets with $b_i = q_i$. Your e-folding rate is now

$$W = \langle \ln(b_i o_i) \rangle = \sum_i p_i \ln \frac{q_i}{r_i} = D(\mathbf{p} \parallel \mathbf{r}) - D(\mathbf{p} \parallel \mathbf{q}) \quad (14.7.19)$$

This will be positive if and only if your estimate of the probabilities is better than the bookie's, that is, closer as measured by the Kullback-Leibler distance. Betting, in other words, is a competition between you and the bookie over who can better estimate the true odds.

A more realistic variant is to assume that the bookie offers payoffs of only some fraction $f < 1$ of his reciprocal probability estimates. Then (you can work out), you can win only if

$$D(\mathbf{p} \parallel \mathbf{r}) - D(\mathbf{p} \parallel \mathbf{q}) > -\ln f \quad (14.7.20)$$

14.7.3 Conditional Entropy and Mutual Information

We now want to look at the association of two variables. Let us return to the guessing game that was discussed in §14.7.1. Suppose we are deciding what question to ask next in the game and have to choose between two candidates, or possibly want to ask both in one order or another. Suppose that one question, x , has I possible answers, labeled by i , and that the other question, y , has J possible answers, labeled by j . Then the possible outcomes of asking both questions form a contingency table whose entries are the joint outcome probabilities p_{ij} , normalized by

$$\sum_{i=0}^{I-1} \sum_{j=0}^{J-1} p_{ij} \equiv \sum_{i,j} p_{ij} = 1 \quad (14.7.21)$$

We use the same “dot” notation as in §14.4 to denote the row and column sums, so that $p_{i\cdot}$ is the probability of outcome i asking question x only, while $p_{\cdot j}$ is the probability of outcome j asking question y only. The entropies of the questions x and y are thus, respectively,

$$H(x) = - \sum_i p_{i\cdot} \ln p_{i\cdot} \quad H(y) = - \sum_j p_{\cdot j} \ln p_{\cdot j} \quad (14.7.22)$$

The entropy of the two questions together is

$$H(x, y) = - \sum_{i,j} p_{ij} \ln p_{ij} \quad (14.7.23)$$

Now what is the entropy of the question y *given* x (that is, if x is asked first)? It is the expectation value over the answers to x of the entropy of the restricted y distribution that lies in a single column of the contingency table (corresponding to the x answer):

$$H(y|x) = - \sum_i p_{i\cdot} \sum_j \frac{p_{ij}}{p_{i\cdot}} \ln \frac{p_{ij}}{p_{i\cdot}} = - \sum_{i,j} p_{ij} \ln \frac{p_{ij}}{p_{i\cdot}} \quad (14.7.24)$$

Correspondingly, the entropy of x given y is

$$H(x|y) = - \sum_j p_{\cdot j} \sum_i \frac{p_{ij}}{p_{\cdot j}} \ln \frac{p_{ij}}{p_{\cdot j}} = - \sum_{i,j} p_{ij} \ln \frac{p_{ij}}{p_{\cdot j}} \quad (14.7.25)$$

We can readily prove that the entropy of y given x is never more than the entropy of y alone, i.e., that asking x first can only reduce the usefulness of asking y (in which case the two variables are *associated*):

$$\begin{aligned} H(y|x) - H(y) &= - \sum_{i,j} p_{ij} \ln \frac{p_{ij}/p_{i\cdot}}{p_{\cdot j}} \\ &= \sum_{i,j} p_{ij} \ln \frac{p_{\cdot j} p_{i\cdot}}{p_{ij}} \\ &\leq \sum_{i,j} p_{ij} \left(\frac{p_{\cdot j} p_{i\cdot}}{p_{ij}} - 1 \right) \\ &= \sum_{i,j} p_{i\cdot} p_{\cdot j} - \sum_{i,j} p_{ij} \\ &= 1 - 1 = 0 \end{aligned} \quad (14.7.26)$$

Quantities like $H(x|y)$ or $H(y|x)$ are called *conditional entropies*. You can easily show that

$$H(x, y) = H(x) + H(y|x) = H(y) + H(x|y) \quad (14.7.27)$$

sometimes called the *chain rule for entropies*. It immediately follows that

$$H(x) - H(x|y) = H(y) - H(y|x) \equiv I(x, y) \quad (14.7.28)$$

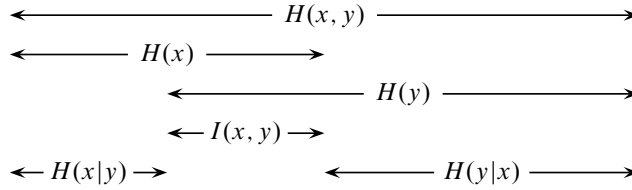


Figure 14.7.2. Relations among the entropies, conditional entropies, and mutual information of two variables. The quantities shown as segment lengths are always positive.

a quantity called the *mutual information* between x and y , given explicitly by

$$I(x, y) = \sum_{i,j} p_{ij} \ln \left(\frac{p_{ij}}{p_{i\cdot} p_{\cdot j}} \right) \quad (14.7.29)$$

Notice that the mutual information is symmetrical, $I(x, y) = I(y, x)$.

Figure 14.7.2 provides a handy way to visualize the additive relations and inequalities among the quantities discussed. As before, all the quantities are measured in bits or nats. Using mutual information, one can make statements like this about the degree of association of two variables: “The variables have information (entropy) 6.5 and 4.2 bits, respectively. However, their mutual information is 3.8 bits, so together they provide only 6.9 bits of information.”

As a more detailed example, let us go back to the racetrack one last time. Suppose that you have some *side information* relevant to the outcome, but not completely predictive. That is, x is the random variable of which outcome i wins, while y is a random variable whose value j you know. Instead of a simple set of probabilities p_i , we now have a contingency table of joint outcomes, p_{ij} . How should you bet, and what is your expected e-folding rate?

First, we need to generalize equation (14.7.17). Suppose that b_{ij} is the fraction of assets that we bet on outcome i when our side information has the value j . There are now J separate constraints,

$$\sum_i b_{ij} = 1, \quad 0 \leq j \leq J-1 \quad (14.7.30)$$

For simplicity, let us take the case where the payoffs are for fair bets (but without the side information), $o_i = 1/p_{i\cdot}$. Then we want to maximize

$$W = \left\langle \ln \frac{b_{ij}}{p_{i\cdot}} \right\rangle = \sum_{i,j} p_{ij} \ln \frac{b_{ij}}{p_{i\cdot}} \quad (14.7.31)$$

A simple calculation, now with J distinct Lagrange multipliers, gives the result,

$$b_{ij} = \frac{p_{ij}}{p_{\cdot j}} \quad (14.7.32)$$

This is again proportional betting, except that it is now conditioned on the value j that is known to us. Substituting equation (14.7.32) into (14.7.31) gives

$$W = \sum_{i,j} p_{ij} \ln \left(\frac{p_{ij}}{p_{i\cdot} p_{\cdot j}} \right) = I(x, y) \quad (14.7.33)$$

We see that the expected e-folding rate is exactly the mutual information between x and y . In other words, we can make money if and only if our side information y has nonzero mutual information with the outcome x . As in equation (14.7.20), you can easily work out more realistic cases where the payouts are not fair bets, or are based on inexact estimates of the true probabilities. A special case of equation (14.7.33) is when the side information y predicts the outcome x *perfectly*. Then, $I(x, y) = H(x) = H(y) = H(x, y)$ and we recover exactly equation (14.7.6).

14.7.4 Uncertainty Coefficients

By analogy with the various coefficients of correlation discussed earlier in this chapter, one sometimes sees *uncertainty coefficients* defined from the various entropies defined above (and in Figure 14.7.2). The uncertainty coefficient of y with respect to x , denoted $U(y|x)$, is defined by

$$U(y|x) \equiv \frac{H(y) - H(y|x)}{H(y)} \quad (14.7.34)$$

This measure lies between 0 and 1, with the value 0 indicating that x and y have no association and the value 1 indicating that knowledge of x completely predicts y . For in-between values, $U(y|x)$ gives the fraction of y 's entropy $H(y)$ that is lost if x is already known. In our game of “twenty questions,” $U(y|x)$ is the fractional loss in the utility of question y if question x is to be asked first.

If we wish to view x as the dependent variable and y as the independent one, then interchanging x and y we can of course define the dependency of x on y ,

$$U(x|y) \equiv \frac{H(x) - H(x|y)}{H(x)} \quad (14.7.35)$$

If we want to treat x and y symmetrically, then the useful combination turns out to be

$$U(x, y) \equiv 2 \left[\frac{H(y) + H(x) - H(x, y)}{H(x) + H(y)} \right] \quad (14.7.36)$$

If the two variables are completely independent, then $H(x, y) = H(x) + H(y)$, so (14.7.36) vanishes. If the two variables are completely dependent, then $H(x) = H(y) = H(x, y)$, so (14.7.35) equals unity. You can easily show that

$$U(x, y) = \frac{H(x)U(x|y) + H(y)U(y|x)}{H(x) + H(y)} \quad (14.7.37)$$

that is, that the symmetrical measure is just a weighted average of the two asymmetrical measures (14.7.34) and (14.7.35), weighted by the entropy of each variable separately.

Generally we find the entropy measures themselves, in bits or nats, more useful than the uncertainty coefficients derived from them.

CITED REFERENCES AND FURTHER READING:

Shannon, C.E., and Weaver, W. 1949, *The Mathematical Theory of Communication*, reprinted 1998 (Urbana, IL: University of Illinois Press).[1]